

Elliot Hong

Ellioth00@berkeley.edu, (424) 394-9378, Berkeley, CA, LinkedIn: <https://www.linkedin.com/in/elliott-hong/>

Security Clearance: Active DoD **Secret level** security clearance (2023 September - Current)

EDUCATION

University of California, Berkeley

Master of Science Major: Mechanical Engineering

Graduated May 2026

Bachelor of Science Major: Mechanical Engineering Minor: Electrical Engineering & Computer Sciences **December 2024**

Relevant Coursework: Operating Systems, Compilers, AI, Autonomous Flight Controls, Microprocessor Design, IoT, Embedded Systems & RTOS, Robotic Locomotion & Control, Deep learning Computer Vision, Digital Design and Integrated Circuits

Technical skills: C/C++, Python, ROS2, OpenCV, FreeRTOS, RTOS (Pintos), MATLAB, Java, Verilog, RISC-V, FPGA (Xilinx), Vitis, MicroPython, Arduino, UART, I2C/SPI, LabVIEW, PCB Design(Eagle), Controls, Signal Processing, SolidWorks, Fusion 360, CST Studio, LPKF, PrusaSlicer, OpenRocket, Power Electronics, Pytorch, Systems Programming, HDL, Memory Management

PROFESSIONAL AND LEADERSHIP EXPERIENCE

UC Berkeley B.E.S.T lab Squishy robotics

January 2024 - Current

- Engineered a PCB using Eagle CAD for a dynamic robot, achieving maneuverability through precise center-of-mass adjustments
- Used Teensy and XBee module for mesh communication between twelve slave machines and one master device
- Created custom PCB boost converter to regulate 2S battery to 12V and tested performance on 10A 7.4V to consistent 12V output

UC Berkeley Student Instructor

April 2023 - May 2025

- Managed an upper division course of 150 students in the utilization of MicroPython to program ESP32 IoT microcontroller and provided comprehensive coverage of complex circuit analysis, phasor, and impedance concepts
- Facilitated lab sessions for 100+ students, providing technical instruction on MATLAB/Python procedural/object-oriented programming, data structures, and the numerical solution of ODEs for engineering analysis

UC Berkeley TAF research lab - Undergraduate researcher

January 2024 - December 2024

- Lead the electric engineering team in autonomous sailboat sensors and actuators such as sail controls and custom made wind vans
- Designed various monohull sailboat lower hull shapes to optimize wave-breaking efficiency and drag force in the ocean

Naval Air Warfare Center Weapons Division - STEM Student Intern

June 2023 - August 2023

- Contributed to the design, development, procurement, and maintenance of U.S. naval aviation platforms by programming the FPGA board using Vitis model composer and matlab to perform anti-aliasing filters and beam streaming using phased arrays
- Designed patch antennas using CST software and manufactured reference loop antennas with the LPKF CNC machine to verify the reliability of anechoic chambers and the new GTEM for future antenna experiments

Station1 Frontiers Internship / Climate Trace - MIT Research fellow

June 2022 - August 2022

- Enhanced global CO2 tracking by integrating real-time data from 400+ power plants using Pandas and AI-driven satellite imagery analysis, presenting findings on automated emission monitoring to 30+ stakeholders

UCLA Summer Scholars Research Internship

June 2021 - August 2021

- Conducted research at the UCLA Nanotechnology Lab, specializing in streamlining the synthesis of gold nanoparticles by consolidating a two-step process into a single step through the use of sodium borohydride

PROJECTS

Custom 3-Stage RISC-V Processor — Xilinx FPGA, Verilog

- Designed and verified a pipelined RISC-V CPU implementing the M-extension with hazard detection and forwarding logic.
- Integrated a UART peripheral for off-chip I/O and validated execution against custom assembly test programs; collaborated via Git on a shared FPGA repository.

V-JEPA Object Tracking — PyTorch, Computer Vision

- Built an object-tracking pipeline in PyTorch using V-JEPA 2.1 with CNN decoders to forecast projectile trajectories behind occlusions via latent-space prediction
- Developed a secondary, standalone U-Net architecture to serve as a performance baseline and reliable fallback, validating the enhanced accuracy of the primary V-JEPA model.

Rocket Attitude Controller — ESP32, FreeRTOS

- Built a roll-axis stabilization controller on ESP32 using Espressif FreeRTOS, with separate real-time tasks for IMU sampling, control loop execution, and telemetry. Conducted HiL testing to validate system performance prior to wind tunnel integration.
- Streamed live attitude and IMU sensor data wirelessly to a LabVIEW GUI for ground-station monitoring.

Pintos Operating system — C, x86 Assembly

- Programmed the Pintos operating system by integrating multithreading, caching, and advanced file system management featuresEngineered semaphores, monitors, and cache structures to enable parallelism, ensuring synchronization and preventing memory overflow in the kernel